

Latest in Post-Quantum Cryptography Migration -- diff report

Generated: 2026-08-29T12:17:05.549496 | **Period:** since 2026-08-29 | **Leaves with news:** 39 of 99

Fresh developments: 32 (as_of within 14 days of the window) | **Superseded/conflicting:** 11 | **Background captured:** 49 (older material, first time in the knowledge base)

Reconfirmations in this period (no new information, not shown): 7.

Summary

Leaf	Fresh	Superseded	Conflicting	Background
Attestation protocols & silicon RoT	2	1	0	1
Public CA ecosystem	2	0	1	0
QRNG & QKD adjacency	3	0	0	1
OTA update signing	2	0	0	4
Language runtime TLS surfaces	0	2	0	2
Enterprise CA platforms	1	1	0	0
IPsec & VPN protocols	2	0	0	3
CDN & edge	2	0	0	0
Key escrow, archive & re-encryption	2	0	0	1
Peripherals & physical access	2	0	0	1
Interop & test tooling	2	0	0	0
Procurement, contract language & audit	2	0	0	0
OpenSSL and forks	0	0	1	1
Linux distributions & kernel crypto	0	1	0	0
Passwordless & hardware tokens	1	0	0	0
Revocation & status infrastructure	0	1	0	0
Long-term signature validation & document signing	1	0	0	0
TLS 1.3 hybrid key exchange	1	0	0	0
QUIC, HTTP/3, DTLS	1	0	0	2
HSM vendors	0	0	1	2
Disk & data-at-rest encryption	1	0	0	0
Automotive	1	0	0	2
Short-range IoT	1	0	0	0
GNSS & satellite links	1	0	0	1
Crypto discovery & CBOM	0	1	0	2

Certification schemes	1	0	0	0
CPU & ISA extensions	1	0	0	0
ERP & business platforms	0	1	0	0
Language-native crypto stacks	0	0	0	2
Embedded & alternative libraries	0	0	0	5
Apple platforms	0	0	0	3
TPM & TCG specifications	0	0	0	2
Artifact signing frameworks	0	0	0	3
Service mesh & workload identity	0	0	0	3
Identity provider platforms	0	0	0	1
SSH	0	0	0	2
DNS & time	0	0	0	2
Performance & side-channel tooling	0	0	0	2
EU & UK guidance	0	0	0	1

Firmware & root of trust

Attestation protocols & silicon RoT (~~attestation-protocols~~)

New developments (2)

SPDM

- DMTF SPDM standards page (updated 2026-08-28) lists SPDM 1.4.1 as published July 2026 and an upcoming SPDM Authorization White Paper targeted for 2026Q3. ([SPDM | DMTF](#), as of 2026-08-28, confidence: high, source: grok)
- *New information about SPDM 1.4.1 publication and an upcoming Authorization White Paper, not previously present.*

measured boot

- TCG issued guidance defining requirements for PQC-ready TPMs, identifying PC Client Platform TPM Profile (PTP) 1.07 as the baseline and establishing separate 'PQC-ready' and 'PQC-upgradable' designations based on TPM 2.0 Library v1.85 PQC support. ([TCG Defines Requirements for PQC-Ready Trusted Platform Modules](#), as of 2026-08-24, confidence: medium, source: grok)
- *New TCG certification/designation framework (PTP 1.07 baseline, PQC-ready vs PQC-upgradable) not previously recorded in the knowledge base.*

Superseded (1)

SPDM

- libspdm reference implementation reached version 3.8.2 as of April 2026. ([SPDM | DMTF](#), as of 2026-08-28, confidence: high, source: grok)

- supersedes: "[d0002:3] libspdm 3.8 was released, including an update to OpenSSL 3.5.1 which adds PQC algorithm support"
- *Reports a newer point release (3.8.2) of libspdm beyond the previously recorded 3.8 release.*

Newly captured background -- older material, first time in the knowledge base (1)

Caliptra

- Caliptra implements a dual-signature scheme using ECDSA P-384 and ML-DSA-87 for authenticating firmware images (including FMC and Runtime layers) and generating dual-signed certificates in its layered DICE identity architecture. ([Caliptra.md](#), as of undated, confidence: high, source: grok)
- *Elaborates on the previously recorded dual ML-DSA-87/ECDSA scheme [d0001:8] with new specifics (P-384 curve, FMC/Runtime firmware layers, dual-signed certificates, DICE identity architecture) not stated before.*

OTA update signing (ota-update-signing)

New developments (2)

Firmware update signing schemes

- Microsoft plans to transition Windows Production signing to PQC-by-default in 2027, following CA replacement in 2026, with considerations for hybrid signatures and legacy systems. ([Preparing the Windows ecosystem for next-generation code signing](#), as of 2026-08-20, confidence: high, source: grok)
- *New vendor-specific migration timeline (Windows code signing) not present in existing knowledge.*

PQ migration paths

- IETF draft-ietf-jose-pq-composite-sigs describes PQ/T hybrid composite signatures for JOSE and COSE, combining ML-DSA with ECDSA or EdDSA for artifact and firmware signing. ([PQ/T Hybrid Composite Signatures for JOSE and COSE](#), as of 2026-08-21, confidence: high, source: grok)
- *Introduces a new composite-signature standardization effort (JOSE/COSE) distinct from the existing non-composite hybrid approach described in the pquip-hsm-constrained draft.*

Newly captured background -- older material, first time in the knowledge base (4)

Firmware update signing schemes

- UEFI Security Sub-Team published a whitepaper outlining PQC updates to the UEFI specification, including support for multiple signatures in PKCS#7 SignedData, TBSCertificate hash in DB, and updates to the UEFI CA for PQC algorithms like ML-DSA and SLH-DSA. ([POST-QUANTUM CRYPTOGRAPHY: UEFI SPECIFICATION UPDATES](#), as of 2026-04-29, confidence: high, source: grok)
- *New detail on UEFI specification-level PQC changes (multi-signature PKCS#7, CA updates) not previously covered in the knowledge base.*
- Western Digital adopted ML-DSA-87 with dual-signing (RSA-3072 hybrid) for secure boot, firmware, and package signing in enterprise HDDs, with PQC PKI/HSM deployed in 2025. ([Post-Quantum Cryptography for Enterprise HDD Security](#), as of 2026-05-15, confidence: medium, source: grok)
- *New vendor deployment example (Western Digital enterprise HDD firmware signing) not previously in the knowledge base.*

OTA update signing

- Android 17 integrates ML-DSA into Android Verified Boot (AVB) for quantum-resistant signatures in the boot sequence and updates Android Keystore and Google Play for ML-DSA app signing. ([Google Online Security](#)

[Blog: March 2026](#), as of 2026-03-31, confidence: high, source: grok)

- *New platform-specific PQC secure-boot/signing adoption (Android AVB, Keystore, Play) not previously recorded.*
- SECQAI demonstrated the first PQC signing (FIPS 204 ML-DSA and FIPS 205 SLH-DSA) in a FOTA update process on its CHERI-based Q-Locked SE01 SoC with a global defence contractor. ([SECQAI and global defence contractor demonstrate world's first PQC in a FOTA process on a Memory Safe Chip](#), as of 2025-11-26, confidence: medium, source: grok)
- *New distinct hardware platform/vendor demonstration of PQC FOTA signing, separate from the existing ESP32-based demonstration.*

TPM & TCG specifications (~~tpm-and-tcg~~)

Newly captured background -- older material, first time in the knowledge base (2)

DICE

- DICE Attestation Architecture V1.3 RC1 and DICE Certificate Profile V1.2 RC1 entered public review with periods from July 23, 2026 to September 21, 2026. ([Public Review Specifications](#), as of 2026-07-23, confidence: high, source: grok)
- *Reports a new draft revision cycle (v1.3 RC1 / Certificate Profile v1.2 RC1) beyond the currently ratified v1.2, not previously recorded.*

TCG specs

- TCG published Platform Certificate Profile Version 2.1 on January 27, 2026, stating that the Root of Trust (RoT) is not limited to the TPM Endorsement Key and can be based on technologies such as DICE or MARS. ([TCG Platform Certificate Profile](#), as of 2026-01-27, confidence: high, source: grok)
- *New specification version broadening acceptable Root-of-Trust technologies beyond the TPM EK; not present in the base snapshot.*

PKI, certificates, identity & authentication

Public CA ecosystem (~~public-ca-ecosystem~~)

New developments (2)

CA/B Forum roadmap

- No new CA/Browser Forum ballot specifically enabling or requiring publicly-trusted PQC TLS server certificates had been adopted or merged into Baseline Requirements as of late August 2026; recent ballots (e.g., SMC017v2 adopted July 30, 2026) addressed RSA key sizes instead. ([Ballot SMC017v2: Increase Minimum RSA CA Key Size | CA/Browser Forum](#), as of 2026-08-27, confidence: medium, source: grok)
- *Introduces a genuinely new, checkable fact — the specific adopted ballot SMC017v2 on RSA CA key sizes as the Forum's most recent activity — not previously recorded, alongside continuing absence of a PQC TLS ballot.*

Public CAs

- DigiCert published updated recommendations for Post-Quantum TLS Migration on August 26, 2026, reiterating ML-DSA-only (not composite) as the default for PQC certificate authentication in public Web PKI contexts and noting that CA/Browser Forum discussions around post-quantum public trust certificates are focused on ML-DSA. ([Recommendations for Post-Quantum TLS Migration | DigiCert](#), as of 2026-08-26, confidence: high,

source: grok)

- *Beyond reiterating the known ML-DSA-only default, this adds the new detail that ongoing CA/Browser Forum discussions on public-trust PQC certificates are specifically centered on ML-DSA, which was not previously recorded.*

Conflicting -- needs review (1)

CT logs

- Chrome Root Program updated its policy page on August 28, 2026, reaffirming that it has no immediate plan to add traditional X.509 certificates containing post-quantum cryptography to the Chrome Root Store and will pursue Merkle Tree Certificates (MTCs) as the primary path, with no planned implementation timelines for MTCs unless explicitly stated. ([Chrome Root Program - Moving Forward, Together](#), as of 2026-08-28, confidence: high, source: grok)
- conflicts with: "[d0002:10] Google's MTC rollout plan includes a Phase 2 targeted for Q1 2027 in which it will invite CT Log operators... to participate in the initial bootstrapping of public MTCs"
- *The August 28 Chrome Root Program policy statement that it has no planned implementation timelines for MTCs sits in tension with the earlier reported concrete Phase 2/Q1 2027 rollout timeline, and it's unclear whether the timeline was walked back or simply omitted here.*

Enterprise CA platforms (enterprise-ca-platforms)

New developments (1)

ADCS

- Microsoft Learn documentation updated 2026-08-25 details AD CS ML-DSA support phases, with Phase 1 (ML-DSA signatures) available and Phase 2 (ML-KEM, composite ML-DSA, composite ML-KEM) planned; requires Windows Server 2025 with KB5087539 or later on CA servers and Windows 11 24H2/25H2 with KB5067036 or later on clients. ([Post-Quantum Cryptography in AD CS overview - Windows Server](#), as of 2026-08-25, confidence: high, source: grok)
- *Adds new specifics — explicit Phase 2 scope (ML-KEM, composite ML-DSA, composite ML-KEM) and the previously unrecorded client-side KB5067036/Windows 11 24H2-25H2 requirement — not present in existing phased-rollout claims.*

Superseded (1)

Venafi

- CyberArk Certificate Manager (Trust Protection Foundation) upgrade notes (Aug 2026) state that experimental PQ keys/certificates are invalidated upon upgrade to versions supporting final FIPS 204/205 standards; PQ algorithms are now available by default without a registry flag. ([Important Considerations before upgrading CyberArk Trust Protection Foundation](#), as of 2026-08-21, confidence: medium, source: grok)
- supersedes: "[d0002:19] Current Venafi documentation labels PQC signing key algorithms as 'Experimental': PQC environments must use software (not HSM) key storage... processing time is higher due to larger key/signature sizes, and PQC support is enabled by default on CodeSign Protect clients."
- *Updates the prior 'Experimental' status: PQ algorithms are now default-enabled without a registry flag and experimental-era PQ keys/certs are invalidated on upgrade to final-standard-supporting versions.*

Passwordless & hardware tokens (passwordless-and-tokens)

New developments (1)

PIV/CAC smart cards

- As of August 27, 2026, GSA is updating the Federal Identity, Credential, and Access Management (FICAM) architecture to support post-quantum cryptography, and its laboratory for testing physical access control systems has begun evaluating quantum-resistant technology, with discovery/planning in 2026-2027, pilots in 2027-2028, and full migration targeted by 2035. ([US begins post-quantum upgrade of federal identity credentials | Biometric Update](#), as of 2026-08-27, confidence: high, source: grok)
- *Adds new detail on GSA's broader FICAM architecture update and physical access control lab testing with a distinct multi-phase timeline through 2035, beyond the previously known Shatterproof Digital Identity pilot.*

Revocation & status infrastructure (revocation-status-infra)

Superseded (1)

Merkle Tree Certificates

- Editor's copy of draft-ietf-plants-merkle-tree-certs (post-v05) as of 2026-08-28 includes renames (e.g., MerkleTreeCertEntry to MTCLogEntry), fixed test vectors, updates to CA certificate representation and subtree proofs, and clarifies CRLs/OCSPs apply unmodified to MTCs. ([Merkle Tree Certificates](#), as of 2026-08-28, confidence: high, source: grok)
- supersedes: "[d0004:5] Editor's copy of draft-ietf-plants-merkle-tree-certs (post-v05) includes renames (e.g., MerkleTreeCertEntry to MTCLogEntry), fixed test vectors, and updates to CA certificate representation and subtree proofs since July 2026."
- *Restates the same editor's-copy update with a fresher date but adds a genuinely new detail — explicit clarification that CRLs/OCSPs apply unmodified to MTCs — so it replaces rather than merely reconfirms the prior entry.*

Long-term signature validation & document signing (long-term-signature-validation)

New developments (1)

ETSI TS 119 312 algorithm catalogue transitions

- ETSI initiated work items for TR 119 330 series on PQC impact on electronic signatures and trust infrastructures (including signature creation/validation, TSPs, and EUDI Wallet), with adoption in June 2026 and early drafts targeted for 2027. ([Work Programme - Work Item Plan](#), as of 2026-08-29, confidence: high, source: grok)
- *New ETSI work-item information about a TR 119 330 series on PQC impacts across signatures/TSPs/EUDI Wallet, not previously recorded in the base or prior deltas.*

Identity provider platforms (idp-platforms)

Newly captured background -- older material, first time in the knowledge base (1)

Keycloak

- A new Keycloak GitHub issue (#50304, opened 2026-06-24, targeting milestone 27.0.0) requests support for PQ/T Hybrid Composite Signatures for JOSE and COSE to enable hybrid signatures combining ML-DSA with ECDSA or EdDSA during the transition to full PQC, referencing the active IETF draft-ietf-jose-pq-composite-sigs. ([Support PQ/T Hybrid Composite Signatures for JOSE and COSE · Issue #50304 · keycloak/keycloak · GitHub](#), as of 2026-06-24, confidence: high, source: grok)

- *This is a distinct, new tracked engineering item (hybrid composite signatures via a separate IETF draft) not previously recorded, complementing but not duplicating the existing ML-DSA-only OIDC signing issue (#50355) and the SAML PQC-readiness issue (#48825).*

HSMs, KMS & key management

QRNG & QKD adjacency (~~qrng-qkd~~)

New developments (3)

QKD developments

- SK Telecom secured Horizon Europe funding as the first Asian private company for a three-year multinational project to develop and demonstrate a next-generation QKD system based on Quantum Photonic Integrated Circuit-Artificial Intelligence (QPIC-AI), involving partners in Greece, Austria, Germany, and Korea. ([SK Telecom Develops Next-Generation Quantum Cryptography Technology with EU Research Funding](#), as of 2026-08-29, confidence: medium, source: grok)
- *New QKD R&D funding and consortium not previously recorded in the knowledge base.*

QRNG developments

- An experimental highly integrated broadband entropy source for QRNG based on vacuum fluctuations, using a hybrid laser-and-silicon-photonics chip, achieved a quantum random number generation rate of 67.9 Gbps under average conditional minimum entropy after digital equalization. ([Highly integrated broadband entropy source for quantum random number generators based on vacuum fluctuations](#), as of 2026-08-24, confidence: medium, source: grok)
- *New QRNG hardware result with a distinct throughput figure and chip design not covered by existing QRNG entries.*
- A tri-type QRNG system was demonstrated that on-demand generates uniform, Gaussian, and Rayleigh random numbers via quantum vacuum noise measurement, achieving over 42 Gbps secure uniform bit rate and over 14 Gbps secure Gaussian bit rate after extraction. ([40Gbps Tri-type Quantum Random Number Generator](#), as of 2026-08-24, confidence: medium, source: grok)
- *New QRNG capability (multi-distribution on-demand generation) not previously present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (1)

QKD developments

- Researchers demonstrated a practical side-channel-secure QKD (SCS-QKD) protocol over 200 km of fiber with secure key rates of 196.03 bps at 200 km, 2.55 kbps at 150 km, and 18.31 kbps at 100 km using a stable 1.25 GHz system and finite-key analysis. ([Proof-of-principle experimental demonstration of side-channel-secure quantum key distribution over 200 km](#), as of 2026-08-11, confidence: high, source: grok)
- *New distinct QKD protocol demonstration and distance/key-rate results not present in existing QKD entries.*

Key escrow, archive & re-encryption (~~key-escrow-archive-recovery~~)

New developments (2)

key custody

- Thales launched the Luna 8 HSM with support for post-quantum cryptography algorithms, an upgradeable architecture for future standards, and targeting PQC readiness for enterprise and government key custody

applications. ([Thales launches Luna 8 hardware security module in India, targeting post-quantum cryptography readiness for enterprises and government](#), as of 2026-08-29, confidence: medium, source: grok)

- *New Thales HSM product (Luna 8) with upgradeable PQC architecture is a distinct fact not covered by the existing Luna HSM firmware-update claim.*
- The Trusted Computing Group released guidance on August 24, 2026, to verify PQC-ready TPMs meeting PTP 1.07 requirements, defining 'TCG PQC-ready TPM' and 'TCG PQC-upgradable TPM' categories for hardware key custody during PQC transitions. ([New Guidance Helps Businesses Verify Quantum-Safe Hardware Claims](#), as of 2026-08-24, confidence: medium, source: grok)
- *Introduces a new TCG certification scheme and categories for TPM key custody not previously present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (1)

archive & re-encryption

- QuCloud proposes a hybrid framework combining E91 QKD, Kyber PKE, SHA-256 hashing, and custom proxy re-encryption for quantum-resistant cloud storage security and mitigation of harvest-now-decrypt-later risks on archived data. ([QuCloud : Enhancing cloud storage security by combining quantum key distribution, post-quantum cryptography, and custom proxy re-encryption](#), as of 2026-06-01, confidence: low, source: grok)
- *New academic proxy re-encryption/QKD hybrid proposal for archived cloud data, not previously recorded under archive & re-encryption.*

HSM vendors (hsm-vendors)

Conflicting -- needs review (1)

Utimaco

- Utimaco Quantum Protect algorithms (ML-KEM, ML-DSA, LMS) passed NIST CAVP validation with specific certificate references; SLH-DSA and additional schemes like XMSS supported via the package. ([Utimacos Nist Validated PQC Algorithms Setting Standard](#), as of 2026-07-31, confidence: medium, source: grok)
- conflicts with: "[d0005:3] Utimaco u.trust General Purpose HSM Se-Series ... supports Quantum Protect PQC extension with ML-KEM, ML-DSA, LMS, XMSS, and SLH-DSA in progress."
- *Claims SLH-DSA is already supported via the package, contradicting the established and more recently reconfirmed status that SLH-DSA remains in progress/on the roadmap.*

Newly captured background -- older material, first time in the knowledge base (2)

Entrust nShield

- Entrust nShield 5s HSM datasheets list specific ML-DSA signing performance (e.g., up to 2,780 TPS for ML-DSA-44 on high model) and support for 18 SLH-DSA variants plus LMS via Post-Quantum Option Pack. ([nShield 5s HSM datasheet](#), as of 2026-06-08, confidence: high, source: grok)
- *Introduces new granular performance and algorithm-variant-count details not previously recorded for nShield 5s.*

Thales

- Thales Luna 8 Network HSM is available now as a network appliance in three models (Luna 100, 150, 190) with a custom Thales cryptographic processor delivering up to 100x faster PQC algorithm performance and an upgradeable architecture for future standards. ([Introducing Luna 8: Our Fastest HSM Ever, Built for the Post-Quantum Era](#), as of 2026-08-04, confidence: high, source: grok)
- *Adds specific model names, current availability status, and a concrete 100x performance figure not present in*

the earlier general Luna 8 launch claim.

Application dev stacks & languages

Language runtime TLS surfaces (~~language-runtimes-tls~~)

Superseded (2)

C/C++

- wolfProvider (July 2026) adds CAVP-validated ML-KEM and ML-DSA (plus hybrids) to OpenSSL 3.x applications via wolfCrypt backend, with no application code changes required. ([wolfProvider Post-Quantum Cryptography: ML-KEM and ML-DSA - wolfSSL](#), as of 2026-07-02, confidence: high, source: grok)
- supersedes: "[d0002:6] wolfProvider (July 2026) adds CAVP-validated ML-KEM/ML-DSA support to OpenSSL 3.x applications via wolfCrypt backend"
- *Same wolfProvider claim now specifies additional detail (hybrid group support and no required application code changes) beyond the original.*

Python

- pyca/cryptography >=48 (released June 2026) adds native support for ML-KEM (FIPS 203) and ML-DSA (FIPS 204) via Rust bindings and AWS-LC backend, making PQC one pip install away. ([Shipping post-quantum cryptography to Python - The Trail of Bits Blog](#), as of 2026-06-30, confidence: high, source: grok)
- supersedes: "[b1:22] pyca/cryptography has an open tracking/roadmap issue (as of April 2026) discussing whether and how to add ML-KEM, ML-DSA, and SLH-DSA support, indicating these algorithms are not yet part of its core API"
- *The roadmap discussion has resolved into shipped native support as of pyca/cryptography 48 (June 2026), superseding the earlier 'not yet part of core API' status.*

Newly captured background -- older material, first time in the knowledge base (2)

C/C++

- wolfSSL 5.9.2 (June 2026) adds ML-DSA (renamed from Dilithium), SLH-DSA, LMS/XMSS support, PKCS11 for ML-KEM, and removes liboqs integrations in favor of native implementations. ([wolfSSL 5.9.2 release blog - wolfSSL](#), as of 2026-06-30, confidence: high, source: grok)
- *New fact about core wolfSSL C library shipping native ML-DSA/SLH-DSA/LMS-XMSS and dropping liboqs, distinct from the earlier Rust-API roadmap claim and not previously recorded in the C/C++ subtopic.*

Go

- Go 1.27 adds a new crypto/mldsa package implementing ML-DSA (FIPS 204) with three parameter sets, plus support in crypto/x509 and crypto/tls for TLS 1.3 signatures. ([Go 1.27 brings generic methods, post-quantum crypto and a new JSON engine - Northeast Times](#), as of 2026-08-02, confidence: medium, source: grok)
- *First mention of ML-DSA signature support (crypto/mldsa, x509/tls integration) in Go, distinct from the existing ML-KEM key-exchange claims.*

Network & transport protocols

IPsec & VPN protocols (~~ipsec-and-vpn~~)

New developments (2)

IKEv2 with RFC 8784 and 9370

- draft-ietf-ipsecme-ikev2-prf-plus-02 (published August 28, 2026) specifies variable-length output PRFs for IKEv2 and explicitly applies to PQC extensions including RFC 8784, RFC 9370, and RFC 9867. ([Use of Variable-Length Output Pseudo-Random Functions \(PRFs\) in the Internet Key Exchange Protocol Version 2 \(IKEv2\)](#), as of 2026-08-28, confidence: high, source: grok)
- *New draft introducing variable-length PRF support tied to PQC IKEv2 extensions, not previously recorded.*

OpenVPN

- Easy-RSA added first-class PQC support (ML-DSA, ML-KEM, SLH-DSA, Falcon, hybrids) via PR #1432, with updates through August 2026, requiring OpenSSL 3.5+ or OQS provider. ([feat\(pqc\): add native Post-Quantum Cryptography support to Easy-RSA](#), as of 2026-08-15, confidence: high, source: grok)
- *New tooling development (Easy-RSA certificate generation) not previously covered in the OpenVPN section.*

Newly captured background -- older material, first time in the knowledge base (3)

IKEv2 with RFC 8784 and 9370

- Palo Alto Networks PAN-OS IKEv2 implementation via RFC 9242 and RFC 9370 supports a broad set of PQC KEMs including ML-KEM (512/768/1024), BIKE (L1/L3/L5), FrodoKEM variants, HQC (128/192/256), NTRU-Prime sntrup761, and Classic McEliece variants. ([How RFC 9242 and RFC 9370 Resist Quantum Computing Threats](#), as of 2026-07-20, confidence: high, source: grok)
- *Adds a specific, previously unstated list of supported PQC KEM algorithms beyond the general agility claim already recorded for PAN-OS.*
- draft-ietf-ipsecme-ikev2-downgrade-prevention reached the RFC Editor queue on July 8, 2026 after IESG review, defining IKE_SA_INIT_FULL_TRANSCRIPT_AUTH to prevent downgrade attacks in hybrid ML-KEM IPsec setups. ([IPsec Downgrade Attack Survives ML-KEM: Cloudflare Ships Authentication Fix](#), as of 2026-07-09, confidence: high, source: grok)
- *New draft and new mechanism (transcript authentication for downgrade prevention) not present in the knowledge base.*

WireGuard

- Surfshark launched post-quantum encryption on WireGuard for macOS, Linux, and Android in January 2026 using PSK-based hybrid handshake with NIST-approved Kyber. ([Can WireGuard Withstand the Quantum Storm?](#), as of 2026-04-26, confidence: medium, source: grok)
- *Adds a new VPN vendor (Surfshark) not previously mentioned among WireGuard PQC deployments.*

TLS 1.3 hybrid key exchange (tls-hybrid-keyex)

New developments (1)

X25519MLKEM768 adoption defaults

- As of August 28 2026, Google Cloud Load Balancing added X25519MLKEM768 support for Application and Proxy Load Balancers on a staged rollout (opt-in from June 2 2026, enabled by default from October 2026, permanently on with no opt-out from October 2027). ([ML-KEM vs X25519 TLS 1.3: 52% Adoption, +1.5ms \[2026\]](#), as of 2026-08-28, confidence: medium, source: grok)
- *New, specific rollout timeline detail for Google Cloud Load Balancing not previously covered (the base only mentions Google Cloud CDN generically).*

QUIC, HTTP/3, DTLS (quic-http3-dtls)

New developments (1)

QUIC

- The ant-quic Rust crate (v0.27.47, August 2026) implements pure post-quantum QUIC transport using only ML-KEM-768 for key exchange and ML-DSA-65 for authentication with no classical fallback. ([ant-quic - crates.io](#), as of 2026-08-24, confidence: high, source: grok)
- *New concrete implementation detail of a pure-PQC QUIC library not previously recorded.*

Newly captured background -- older material, first time in the knowledge base (2)

DTLS

- RFC 10024 (published August 2026) defines the hybrid groups X25519MLKEM768, SecP256r1MLKEM768, and SecP384r1MLKEM1024 for TLS 1.3 and registers them with DTLS-OK: Y in the IANA TLS Supported Groups registry. ([RFC 10024 - Post-Quantum Traditional \(PQ/T\) Hybrid Key Agreement Mechanisms for TLS 1.3](#), as of 2026-08-01, confidence: high, source: grok)
- *This is a concrete, newly published RFC that formally registers specific hybrid groups with DTLS-OK: Y, going beyond the general framework guidance already known from RFC 9954.*

QUIC

- A December 2025 comparative study found that QUIC reduces the performance penalty (handshake latency and bandwidth overhead) of hybrid and pure post-quantum KEMs compared to TLS, especially under lossy network conditions. ([Comparative Analysis of Post-Quantum Handshake Performance in QUIC and TLS Protocols](#), as of 2025-12-01, confidence: high, source: grok)
- *New quantitative comparative finding about QUIC vs TLS PQ handshake performance not previously present in the knowledge base.*

SSH (ssh)

Newly captured background -- older material, first time in the knowledge base (2)

OpenSSH

- OpenSSH 10.5 (released 2026-08-11) includes an upstream performance improvement for keying the sntrup761 key agreement algorithm. ([OpenSSH: Release Notes](#), as of 2026-08-11, confidence: high, source: grok)
- *New detail identifying the sntrup761 keying performance improvement as an upstream OpenSSH 10.5 change, distinct from the previously noted Debian 1:10.4p1 package-level improvement.*

mlkem768 keyex

- OpenSSH briefly implemented mlkem768brainpoolp256r1-sha256 (added July 2026) before removing it in August 2026 (commit 05ec77c14b9143b4d49f05a8dcd1366a92b9024e). ([OpenSSH: ChangeLog](#), as of 2026-08-08, confidence: high, source: grok)
- *Genuinely new fact about a short-lived key exchange algorithm addition and removal not present anywhere in the existing knowledge base.*

DNS & time (dns-and-time)

Newly captured background -- older material, first time in the knowledge base (2)

DNSSEC

- New IETF Internet-Draft draft-johani-dnsop-dnssec-alg-split-02 (published July 2026) proposes Algorithm-Split DNSSEC, separating large post-quantum KSK algorithms from small-signature ZSK algorithms to mitigate signature size issues in DNSSEC responses. ([draft-johani-dnsop-dnssec-alg-split-02 - Algorithm-Split DNSSEC: KSK/ZSK Algorithm Separation](#), as of 2026-07-03, confidence: high, source: grok)
- *Introduces a distinct new IETF proposal (KSK/ZSK algorithm split) not previously recorded, addressing signature size via a different mechanism than the existing dual-algorithm or MTL approaches.*
- New IETF Internet-Draft draft-kaizer-dnsop-ml-dsa-mtl-dnssec-01 (published August 2026) specifies ML-DSA-MTL (Module-Lattice-Based Signatures with Merkle Tree Ladders) for DNSSEC, applying MTL mode to ML-DSA-44 with SHAKE-128 to address large signature sizes. ([draft-kaizer-dnsop-ml-dsa-mtl-dnssec-01 - Module-Lattice-Based Signatures with Merkle Tree Ladders \(ML-DSA-MTL\) for DNSSEC](#), as of 2026-08-14, confidence: high, source: grok)
- *Provides a concrete new specification applying MTL mode to ML-DSA-44, a specific algorithm/draft combination not previously named (existing knowledge only mentioned MTL mode generically for SLH-DSA, Falcon, XMSS, LMS).*

Cloud infrastructure

CDN & edge (cdn-edge)

New developments (2)

Cloudflare

- As of August 28, 2026, Cloudflare's PQC in Cloudflare products documentation confirms key agreement is enabled with X25519MLKEM768 for both visitor-to-Cloudflare and Cloudflare-to-origin connections, signatures are enabled with ML-DSA for Cloudflare-to-origin via AOP/COTS but only planned via Merkle Tree Certificates for visitor-to-Cloudflare connections, and internal network key agreement is mostly migrated while internal signatures are not yet implemented. ([PQC in Cloudflare products · Cloudflare SSL/TLS docs](#), as of 2026-08-28, confidence: high, source: grok)
- *Adds the previously unrecorded detail that visitor-to-Cloudflare signature authentication is planned via Merkle Tree Certificates, beyond what prior claims about origin authentication and internal migration status covered.*

Fastly

- Fastly, together with Cloudflare, accounts for nearly 70% of all observed PQ-TLS deployments on the internet as of measurement rounds through March 2026. ([Mind the Gap: Policy vs Reality in Post-Quantum TLS Deployment](#), as of 2026-08-24, confidence: medium, source: grok)
- *Introduces a new market-share statistic on PQ-TLS deployment share not previously present in the knowledge base.*

Enterprise infrastructure & storage

Peripherals & physical access (peripherals-physical-access)

New developments (2)

badge and physical access systems

- GSA is updating the FICAM architecture to support post-quantum cryptography with an emphasis on crypto

agility for compatibility with existing systems, alongside expanding the FIPS 201 Evaluation Program / PACS lab to test quantum-resistant solutions. ([GSA Leads the Transition to Quantum-Resistant Technology | GSA](#), as of 2026-08-24, confidence: high, source: grok)

- *Adds a new architectural detail (FICAM modernization emphasizing crypto agility) beyond the previously recorded PACS lab expansion.*
- GSA's FICAM modernization includes an interagency working group whose first meeting occurred on August 12, 2026, with a phased timeline: discovery/planning 2026-2027, pilots 2027-2028, prioritized migration through 2030, post-quantum digital signatures for high-value systems by 2031, and full migration by 2035. ([US begins post-quantum upgrade of federal identity credentials | Biometric Update](#), as of 2026-08-27, confidence: high, source: grok)
- *Provides new concrete timeline and working-group details not present in any existing GSA-related claim.*

Newly captured background -- older material, first time in the knowledge base (1)

badge and physical access systems

- Swissbit iShield Key 2 integrates MIFARE DESFire EV3 credential technology (via NXP SmartMX3 P71D600 MCU) alongside FIDO2 for unified logical and physical access control in a single device, available as of June 2026. ([Swissbit combines FIDO and physical access control in single security key with NXP | Biometric Update](#), as of 2026-06-12, confidence: high, source: grok)
- *Describes a distinct additional credential-technology integration (MIFARE DESFire EV3/NXP) on the iShield Key 2 not covered by the earlier HID Seos integration claim.*

Disk & data-at-rest encryption (disk-and-data-encryption)

New developments (1)

SEDs/TCG Opal

- TCG released guidance (August 2026) defining requirements for PQC-ready TPMs, identifying PTP 1.07 as baseline and designating 'TCG PQC-ready TPM' (implements PTP 1.07) vs 'TCG PQC-upgradable TPM' categories. ([TCG Defines Requirements for PQC-Ready Trusted Platform Modules](#), as of 2026-08-24, confidence: high, source: grok)
- *This is a distinct TCG standard (TPM/PTP) from the Opal spec covered in the existing claim, and introduces a genuinely new PQC-readiness classification not previously recorded in this leaf.*

Developer & migration tooling

Interop & test tooling (interop-test-tooling)

New developments (2)

Interop test servers

- A new free online tool at <https://www.mylesagnew.com/pqctest/> performs a real post-quantum TLS 1.3 handshake against any public HTTPS server on port 443 to test support specifically for the hybrid group X25519MLKEM768 (ML-KEM-768 per FIPS 203). ([Post-Quantum Server Test | Myles Agnew](#), as of 2026-08-25, confidence: high, source: grok)
- *A newly identified interop test tool distinct from existing entries (OQS server, pqcscan), not previously recorded.*

tls-scan

- The openshift/tls-scanner Go module (for OpenShift/Kubernetes TLS compliance scanning) incorporates post-quantum (ML-KEM) readiness checks via testssl.sh, with a dedicated -pqc-check flag that exits non-zero on failure to detect TLS 1.3 + ML-KEM support. ([tls-scanner module - github.com/openshift/tls-scanner](https://github.com/openshift/tls-scanner), as of 2026-08-25, confidence: high, source: grok)
- *New tooling detail about a distinct tls-scan-related project integrating PQC checks via testssl.sh, not previously present in the knowledge base.*

Crypto discovery & CBOM (crypto-discovery-cbom)

Superseded (1)

SBOM integration

- CISA published the final 2026 Minimum Elements for SBOMs on July 29, 2026, adding requirements for SBOM author digital signature, component license metadata, and restricting machine-processable formats to SPDX and CycloneDX only (dropping SWID). ([CISA's 2026 SBOM Minimum Elements Now Require Post-Build Data - OPSWAT](#), as of 2026-07-29, confidence: high, source: grok)
- supersedes: "[d0002:5] CISA's 2026 SBOM Minimum Elements can be satisfied by CycloneDX 1.6 (with exact property paths mapped for each field)"
- *Confirms finalization date and adds concrete new requirements (digital signature, license metadata, dropping SWID) beyond the prior mapping-focused claim.*

Newly captured background -- older material, first time in the knowledge base (2)

Crypto discovery & CBOM

- sbomify v26.7.0 (July 2026) added native CBOM support with NIST-grounded PQC readiness classification and post-quantum posture cards for components. ([Announcing sbomify v26.7.0: The One That Gets Quantum-Ready | Sbomify](#), as of 2026-07-07, confidence: medium, source: grok)
- *Identifies a new tool with CBOM/PQC readiness features not previously recorded.*

SBOM integration

- CISA's finalized 2026 SBOM Minimum Elements emphasize post-build data such as executable artifact hashes and require scans of binaries/container layers rather than source dependencies alone. ([CISA's 2026 SBOM Minimum Elements Now Require Post-Build Data - OPSWAT](#), as of 2026-07-29, confidence: high, source: grok)
- *Introduces a new scope requirement (post-build/binary artifact data) not previously captured in the knowledge base.*

Performance & side-channel tooling (perf-sidechannel)

Newly captured background -- older material, first time in the knowledge base (2)

constant-time verification

- liboqs 0.16.0 release (July 2026) removed the legacy HQC constant-time test as part of updates to algorithm support and testing infrastructure. ([liboqs/RELEASE.md](#), as of 2026-07-09, confidence: high, source: grok)
- *New, specific fact about a liboqs release change to constant-time testing infrastructure not covered by the existing LFX mentorship claim.*

fuzzing

- A May 2026 master's thesis from University of Oulu extends Cryptofuzz with PQC-specific support for differential fuzzing of ML-KEM and ML-DSA implementations across libraries including Botan, OpenSSL, and liboqs, executing ~106M ML-KEM and ~26M ML-DSA test cases. ([A DIFFERENTIAL FUZZING FRAMEWORK FOR ML-KEM AND ML-DSA IMPLEMENTATIONS IN CRYPTOGRAPHIC LIBRARIES](#), as of 2026-05-01, confidence: medium, source: grok)
- *New fuzzing tooling development for ML-KEM/ML-DSA not previously recorded (existing fuzzing entry only covers HQC differential fuzzing by Quarkslab).*

Governance, mandates & timelines

Procurement, contract language & audit (procurement-contract-language)

New developments (2)

RFP and supplier questionnaire language

- On August 28, 2026, CISA's lead for quantum security stated that agencies will modify contract language to require vendors to provide quantum-resistant products, demonstrate PQC compliance pathways, and support mechanisms such as cryptographic bills of materials. ([Quantum security 'drumbeat' to grow louder in federal procurement](#), as of 2026-08-28, confidence: high, source: grok)
- *New concrete statement from a CISA official confirming imminent contract-language changes requiring vendor CBOM support, beyond the general prediction in existing claims.*
- On August 28, 2026, the Department of Defense issued an RFI seeking industry input on software-defined encryption solutions providing packet-level cryptographic protection aligned with PQC migration timelines through December 31, 2031, with responses due September 27, 2026. ([DOD probes industry about software-defined encryption as U.S. scurries to get quantum-safe](#), as of 2026-08-28, confidence: high, source: grok)
- *New, distinct DoD procurement RFI on software-defined encryption with its own timeline, not previously recorded.*

Certification schemes (certification-schemes)

New developments (1)

FIPS 140-3 module transition timelines

- Sansec HSM Cryptographic Module received FIPS 140-3 Level 3 certification (certificate #5503) as the first domestic Chinese HSM supporting the full NIST PQC suite (ML-DSA, ML-KEM, SLH-DSA) alongside classical algorithms. ([三未信安取得FIPS 140-3 Level 3认证，全球化布局再落关键一子](#), as of 2026-08-27, confidence: high, source: grok)
- *New distinct vendor/product (Sansec, China's first) achieving FIPS 140-3 Level 3 with full PQC suite, not previously recorded — existing d0004:1 covers Crypto4A's HSM, a different product.*

EU & UK guidance (eu-uk-guidance)

Newly captured background -- older material, first time in the knowledge base (1)

EU/ETSI roadmap

- As of July 2026, ETSI CYBER QSC work programme includes multiple active draft items such as TR 104 317 on Quantum-safe signatures and PKI considerations (early draft July 2026), TS 104 145 on Quantum-Safe Enterprise Transport Security (final draft July 2026), and DMI/CYBER-QSC-0010 CYBER QSC Extended

Roadmap (stable draft June 2026). ([Work Programme - Work Item Plan](#), as of 2026-07-28, confidence: high, source: grok)

- *Introduces specific new ETSI CYBER QSC work-item details (TR 104 317, TS 104 145, DMI/CYBER-QSC-0010) not previously recorded in the knowledge base.*

Crypto libraries & providers

OpenSSL and forks (**openssl-and-forks**)

Conflicting -- needs review (1)

BoringSSL

- BoringSSL added ML-DSA signature support (ML-DSA-44/65/87) around April 2026. ([Cryptography | Noise](#), as of 2026-04-01, confidence: medium, source: grok)
- conflicts with: "[d0002:1] BoringSSL added ML-DSA support including ML-DSA-87 for X.509 certificate validation and OIDs/NIDs for ML-DSA and ML-KEM via commits in October 2025"
- *This claim dates BoringSSL's ML-DSA addition to around April 2026, contradicting the established October 2025 commit dates without clearly superseding them.*

Newly captured background -- older material, first time in the knowledge base (1)

s2n-tls

- s2n-tls's default_pq policy that includes ML-DSA support is specifically version 20250721. ([Post Quantum Support](#), as of 2026-06-15, confidence: high, source: grok)
- *Adds a previously unrecorded specific policy version number to the existing general claim about s2n-tls ML-DSA/default_pq support.*

Language-native crypto stacks (**language-native-crypto**)

Newly captured background -- older material, first time in the knowledge base (2)

.NET CNG

- In .NET 10, MLDsa and SlhDsa members were renamed from *SecretKey* to *PrivateKey* (e.g., ImportMLDsaPrivateKey, PrivateKeySizeInBytes) as a breaking change for alignment with other asymmetric types. ([Breaking change - MLDsa and SlhDsa 'SecretKey' members renamed](#), as of 2026-07-31, confidence: high, source: grok)
- *New API-naming detail about .NET 10 PQC types not previously captured in the knowledge base.*

Java JCA + BouncyCastle

- Bouncy Castle Java 1.85 (July 28, 2026) adds support for hybrid certificates, post-quantum keys and signatures in X.509, and standards-based PKI migration mechanisms. ([New Release: Bouncy Castle Java 1.85](#), as of 2026-07-28, confidence: medium, source: grok)
- *Specifies concrete new PKI/X.509 capabilities in 1.85 beyond the vague 'significant advances' summary already recorded.*

Embedded & alternative libraries (**embedded-alt-libraries**)

Newly captured background -- older material, first time in the knowledge base (5)

GnuTLS

- GnuTLS 3.8.13 (released April 29, 2026) added support for deriving an ML-DSA public key from an expanded private key per RFC 9881. ([Blame · NEWS · master · gnutls / GnuTLS · GitLab](#), as of 2026-04-29, confidence: medium, source: grok)
- *Base snapshot only notes 3.8.13's release date via Wikipedia; this adds a specific new feature (RFC 9881 ML-DSA public-key derivation) not previously recorded.*

wolfSSL

- wolfSSL 5.9.2 (released June 30, 2026) renamed its post-quantum signature implementation from Dilithium to ML-DSA, updated SLH-DSA sign/verify APIs to require pre-hashed message digests, added RFC 9802 LMSS/XMSS support in X.509 certificate and CSR generation, and added ML-KEM support for PKCS#11. ([wolfSSL 5.9.2 release blog - wolfSSL](#), as of 2026-06-30, confidence: medium, source: grok)
- *New release-specific details (API renames, RFC 9802 cert/CSR support, PKCS#11 ML-KEM) not present in the base snapshot.*
- wolfPKCS11 2.1.0 (released July 2, 2026) added post-quantum crypto support for ML-DSA (FIPS 204) and ML-KEM (FIPS 203). ([wolfPKCS11 2.1.0 is out! \(X post\)](#), as of 2026-07-02, confidence: medium, source: grok)
- *Genuinely new claim about a specific wolfPKCS11 release adding ML-DSA/ML-KEM, not previously recorded (blocked retrieval, so added rather than used to supersede anything).*
- wolfSSL added PQC support (ML-KEM, ML-DSA, SLH-DSA, LMS, XMSS) to its Java providers wolfJSSE and wolfJCE, including PQC-enabled TLS 1.3 hybrid key exchange. ([Java is now #PostQuantum ready with wolfSSL! \(X post\)](#), as of 2026-08-11, confidence: medium, source: grok)
- *New information about PQC reaching wolfSSL's Java providers, not covered by any existing base claim.*
- wolfSSL enabled post-quantum and hybrid TLS 1.3 key exchange (including multiple SecPMLKEM and XMLKEM groups) when using curl built with wolfSSL. ([Post-Quantum Cryptography in curl - wolfSSL](#), as of 2026-06-16, confidence: medium, source: grok)
- *New integration detail (curl + wolfSSL PQC hybrid groups) absent from the base snapshot.*

Operating systems & endpoints

Linux distributions & kernel crypto (**linux-distros-kernel**)

Superseded (1)

SUSE

- SUSE Linux Enterprise Server 16.0 enables post-quantum key exchange algorithms by default in system-wide cryptographic policies for SSH connections, re-enabling the X25519 group to support hybrids such as mlkem768x25519-sha256 and sntrup761x25519-sha512. ([SUSE Linux Enterprise Server Release Notes | SUSE Linux Enterprise Server 16.0](#), as of 2026-08-19, confidence: medium, source: grok)
- supersedes: "[d0001:16] The OpenSSH 10.x hybrid ML-KEM/X25519 key exchange was, as of that SUSE analysis, not yet available in SUSE Linux Enterprise products."
- *SLES 16.0 now ships with PQC hybrid SSH key exchange enabled by default via updated crypto policies, superseding the earlier finding that this was not yet available.*

Apple platforms (**apple-platforms**)

Newly captured background -- older material, first time in the knowledge base (3)

iOS

- Quantum-secure TLS is enabled by default in URLSession and Network framework for all system services and apps on iOS 26, iPadOS 26, macOS 26, tvOS 26, and watchOS 26. ([Apple Platform Security Guide](#), as of 2026-08-03, confidence: high, source: grok)
- *New detail that quantum-secure TLS is on-by-default at the API level for system services and third-party apps, going beyond the earlier ClientHello-advertisement claim.*

macOS

- In macOS 26, the SSH protocol was upgraded with a quantum-secure encryption key exchange. ([Apple Platform Security Guide](#), as of 2026-08-03, confidence: high, source: grok)
- *New, specific, Apple-primary-sourced detail confirming SSH itself gained a quantum-secure KEX in macOS 26 -- distinct from and more precise than the earlier contested, PQ3-branded TLS/VPN/SSH claim.*

watchOS

- Quantum-secure encryption between iPhone (iOS 26+) and Apple Watch (watchOS 26+) uses additional ML-KEM-768 key exchanges in addition to elliptic-curve Diffie-Hellman. ([Apple Platform Security Guide](#), as of 2026-08-03, confidence: high, source: grok)
- *Genuinely new detail about a specific iPhone-Watch pairing protocol using hybrid ML-KEM-768 + ECDH, not previously recorded.*

Enterprise IoT & OT

Automotive (automotive)

New developments (1)

V2X

- PQAKA, a post-quantum authentication and key agreement protocol for 5G-V2X using ML-KEM, achieves mutual authentication and forward secrecy while maintaining compatibility with existing 3GPP entities and deployability on USIM/HSM/TEE platforms. ([Post-Quantum Authentication and Key Agreement Protocol for 5G-V2X Communication](#), as of 2026-08-19, confidence: medium, source: grok)
- *Introduces a new named protocol (PQAKA) and technical approach for 5G-V2X PQC authentication not present in the existing knowledge base.*

Newly captured background -- older material, first time in the knowledge base (2)

UNECE R155/156

- Using deprecated classical asymmetric algorithms may risk failing Type Approval under UNECE R155. ([On-demand webinar: Adoption of post-quantum cryptography in real-time automotive systems](#), as of 2026-03-10, confidence: medium, source: grok)
- *Adds a specific new risk linkage (algorithm deprecation to type-approval failure) not stated in existing R155 claims about CSMS/SUMS evidence requirements.*

V2X

- Post-quantum cryptography for intelligent transportation systems: An implementation-focused review (Vehicular Communications, Volume 60, August 2026) identifies thirteen research gaps in PQC adoption for in-vehicle networks and V2X, including non-PQC-ready standards and certificate-management barriers. ([Post-quantum cryptography for intelligent transportation systems: An implementation-focused review](#), as of 2026-08-01, confidence: high, source: grok)

- Same source article as [d0001:7] but surfaces a distinct, previously unrecorded finding (thirteen research gaps in PQC adoption for ITS/V2X) beyond the AUTOSAR crypto-stack detail already captured.

Short-range IoT (**short-range-iot**)

New developments (1)

BLE

- NIST-standardized ML-KEM, ML-DSA, and SLH-DSA parameter sets all execute successfully on ESP32-C6 microcontroller without external memory, with measurements of latency, key/signature sizes, and memory use under embedded constraints including BLE telemetry scenarios. ([Exploring hardware implementation feasibility of post-quantum cryptography in embedded systems: Evaluation of NIST-standardized ML-KEM, ML-DSA and SLH-DSA on ESP32-C6](#), as of 2026-08-18, confidence: medium, source: grok)
- *New empirical evidence that all three ratified NIST PQC algorithms run successfully on a specific low-power microcontroller in BLE telemetry contexts, not previously recorded in the knowledge base.*

GNSS & satellite links (**gnss-satellite**)

New developments (1)

satcom link encryptors

- WISeKey subsidiary WISeSat.Space signed a strategic cooperation agreement with Kaynes Space Technology (KSTPL) to manufacture post-quantum-secure WISeSat satellites in India, integrating SEALSQ post-quantum cryptographic semiconductors and WISeKey Root of Trust, establishing India as an official launch location alongside the US. ([WISeKey, Sealsq team with KSTPL to manufacture post-quantum satellites in India](#), as of 2026-08-27, confidence: medium, source: grok)
- *New manufacturing/partnership development for PQC-secured satellites not previously recorded, distinct from the existing SEALSQ QSOC launch claim.*

Newly captured background -- older material, first time in the knowledge base (1)

satcom link encryptors

- Voyager Technologies and IBM demonstrated the first post-quantum-secured communication link between the International Space Station and Earth in April 2026, using IBM Quantum Safe Remediator (with NIST PQC standards ML-KEM/FIPS 203, ML-DSA/FIPS 204, SLH-DSA/FIPS 205) deployed on Voyager's Space Edge micro datacenter as a proxy for legacy applications without code changes. ([Voyager and IBM Prove Post-Quantum Security On-Orbit](#), as of 2026-04-15, confidence: high, source: grok)
- *New, corroborated demonstration of a PQC-secured ISS-Earth link, distinct from the previously recorded QuSecure Starlink demonstration.*

Hardware acceleration & silicon

CPU & ISA extensions (**cpu-isa-extensions**)

New developments (1)

vendor-optimized PQ assembly upstreamed into libraries

- wolfSSL retired its remaining dependency on liboqs in August 2026, completing the transition to fully native implementations of ML-KEM, ML-DSA, and other PQC algorithms with optimized paths for x86-64 and AArch64. ([wolfCrypt Retires liboqs as Native Post-Quantum Support Expands](#), as of 2026-08-19, confidence: high, source: grok)
- *New information about wolfSSL fully decoupling from liboqs and shipping native optimized PQC assembly, not previously recorded (prior claim only noted wolfSSL's own AVX2 ML-KEM assembly without addressing liboqs dependency status).*

Enterprise applications & delivery

ERP & business platforms (erp-business-platforms)

Superseded (1)

Oracle Database

- Official Oracle Database documentation as of August 2026 confirms that PQC algorithms (ML-KEM, ML-DSA, SLH-DSA) are not available in FIPS 140-2 mode regardless of cryptographic provider, available only in non-FIPS or FIPS 140-3 mode with the next-generation provider. ([Post-Quantum Cryptography in Oracle Database](#), as of 2026-08-25, confidence: high, source: grok)
- supersedes: "[d0004:3] a August 2026 Oracle blog states the new cryptographic provider in Oracle Database 19.32 supports non-FIPS, FIPS 140-2, and FIPS 140-3 modes"
- *Official Oracle documentation (higher authority than the blog post) clarifies that PQC remains unavailable in FIPS 140-2 mode regardless of provider, resolving the prior unresolved conflict in favor of the original claim.*

Software supply chain

Artifact signing frameworks (artifact-signing-frameworks)

Newly captured background -- older material, first time in the knowledge base (3)

Sigstore/cosign

- As of March 2026, a community proof-of-concept demonstrates end-to-end ML-DSA-65 signing and verification in cosign (import-key-pair, sign-blob, verify-blob) using a patched Go 1.26.0-test build with crypto/mldsa support. ([cosign with post-quantum MLDSA sample · sigstore/cosign · Discussion #4771](#), as of 2026-03-16, confidence: medium, source: grok, kind: chatter, status: unconfirmed, corroboration: single-source)
- *This is a distinct community demonstration using a patched Go stdlib build (crypto/mldsa, import-key-pair) rather than the Trail of Bits custom Go/Rekor implementation already recorded, adding a genuinely new data point on cosign PQC feasibility.*
- Sigstore's algorithm registry currently lists ml-dsa-44, ml-dsa-65, and ml-dsa-87 as experimental pure variants that are not yet fully functional within Sigstore. ([architecture-docs/algorithm-registry.md at main · sigstore/architecture-docs](#), as of undated, confidence: medium, source: grok)
- *Provides new specificity on which ML-DSA parameter sets are registered and their current experimental/non-functional status, a detail not previously captured about the algorithm registry.*

TUF/Notary

- A November 2025 peer-reviewed paper details a phased PQC migration for TUF role hierarchies (root, targets, snapshot, timestamp) and Notary v2 signer certificates, starting with hybrids (RSA/EC + SPHINCS+/Dilithium/

Falcon) then moving to pure PQC algorithms with specific signature size deltas. ([Quantum-Resilient PKI for Container Supply Chains: Post-Quantum Cryptography in TUF and Notary Frameworks](#), as of 2025-11-01, confidence: high, source: grok, kind: analysis)

- *Introduces a new, previously unrecorded phased migration proposal spanning both TUF and Notary v2 together, with concrete hybrid-then-pure algorithm staging not covered by existing TUF-only or Notary-only claims.*

Containers, orchestration, service mesh

Service mesh & workload identity (~~mesh-workload-identity~~)

Newly captured background -- older material, first time in the knowledge base (3)

Consul

- As of July 16, 2026, an open GitHub issue requests that Consul add support for post-quantum algorithms such as X25519MLKEM768 for TLS key exchange, noting that Go's crypto/tls has supported them since version 1.24 but documentation is unclear. ([Support of post-quantum algorithms · Issue #23738 · hashicorp/consul](#), as of 2026-07-16, confidence: medium, source: grok)
- *New, distinct issue about Consul's own TLS stack lacking PQC support, separate from the previously known stale upstream Envoy issue.*

Istio

- As of May 16, 2025, an open Istio GitHub issue requests extending PQC support (X25519MLKEM768) from gateways to mesh-internal traffic including sidecars, waypoints, ztunnel, and xDS, suggesting a new 'pqc' compliance policy. ([Feature request: PQC for mesh-internal traffic · Issue #56330 · istio/istio](#), as of 2025-05-16, confidence: medium, source: grok)
- *Adds new detail about a pre-1.27 feature request explicitly calling out sidecars and waypoints, coverage not confirmed by the existing 1.27 compliance-policy claim.*

SPIFFE/SPIRE

- SPIRE 1.14.3 (March 18, 2026) updated the RequirePQKEM TLS policy to use the standardized X25519MLKEM768 instead of the draft x25519Kyber768Draft00. ([spire/CHANGELOG.md at main · spiffe/spire](#), as of 2026-03-18, confidence: high, source: grok)
- *New fact about an earlier SPIRE release (1.14.3) migrating from draft to standardized KEM naming, not previously recorded.*